12TH CHEMISTRY · CHAPTER 25 · CHAPTER TEST

Organic Synthesis

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 25	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The fragments obtained after disconnection are called:

(a) Products (b) Reagents
(c) Isomers (d) Synthons

2. The reagent which converts a carboxylic acid into an acid halide is:

(a) $NaBH_4$ (b) $SOCl_2$
(c) PBr_3 (d) $LiAlH_4$

3. The synthetic equivalent of the nitronium ion is:

(a) Nitric acid (b) Benzene
(c) Ammonia (d) Aniline

4. Acetophenone is prepared from benzene and:

(a) Acetic acid (b) Ethanol
(c) Acyl chloride (d) Chloromethane

5. The starting material for both aspirin and paracetamol is:

(a) Benzene (b) Phenol
(c) Toluene (d) Aniline

6. Halicin was discovered by researchers at:

(a) Oxford (b) Harvard
(c) MIT (d) Cambridge

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Differentiate between synthesis and retrosynthesis.
ii. What is a synthetic equivalent?
iii. What is functional group interconversion?
iv. Why is an FGI needed in the retrosynthesis of aniline?
v. How is benzoic acid prepared from benzene?
vi. What is Halicin?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Construct the retrosynthetic tree of paracetamol and design its synthesis.
(b) Construct the retrosynthetic tree of aspirin and design its synthesis.

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